The Sentinel Ecosystem

Hardware-Software Infrastructure for Clinical Resilience

Systems Engineering Proposal - Bridging the Psychiatric Monitoring Gap

1. THE PROBLEM - A SYSTEM UNDER STRAIN

Mental healthcare runs on a reactive model. A patient shows up for their weekly session, talks for 50 minutes, and then goes back into the world for another 167 hours with nobody watching. That is where things fall apart. Crises develop between the cracks. Tasks go undone. Patients sit alone with their thoughts and no feedback loop.

On the other side of the desk, the numbers are worse. India runs 0.75 psychiatrists per 100,000 people. The WHO says you need at least 1 per 10,000. That is a 13x gap. Clinicians carry caseloads nobody designed them to handle, drowning in paperwork, progress notes, and the emotional weight of everyone elses trauma. Secondary stress is baked into the job.

You end up with a dual failure: patients lack any continuous support layer, and clinicians burn out from administrative overload. Nobody wins.

Existing solutions do not fix this. Telehealth platforms give you appointment-only access. Crisis helplines average 3-5 minute wait times - forever when you are in distress. Wellness apps are generic and disconnected from the clinical loop. None of them connect patient, psychologist, and trusted contact in a single real-time system. None of them combine what your body is saying with what you wrote in your journal.

2. THE SENTINEL ECOSYSTEM - NOT A PROJECT, AN INFRASTRUCTURE

Sentinel is a hardware-software infrastructure, not another app. Three things make it different from anything out there.

Tri-directional stakeholder loop. Most platforms serve one person. Sentinel serves three - patient, psychologist, and trusted contact - through synchronized interfaces. A patient writes a journal entry. The psychologist sees the summary on their triage board. If the crisis engine fires, the trusted contact gets an alert too. Everyone sees the same reality at the same time.

Hardware-software integration. Biometric data from a wearable ring feeds into the same assessment pipeline as journal text. Heart rate variability, sleep patterns, stress levels - analyzed alongside subjective mood and written reflection. One unified stream instead of siloed dashboards.

End-to-end workflow closure. Crisis trigger to acknowledgment. Task assignment to graded feedback. Booking submission to status notification. Every loop closes inside the ecosystem. Nothing falls out.

2.1 Patient Portal

Patients get a daily journal with AI summarization that works in two modes: a warm personal reflection for them, and a structured clinical OAP note for their psychologist. Mood tracking locks one entry per day - no retrospective filling. Emotion classification covers 28 GoEmotions labels (sadness, fear, nervousness, etc.) feeding into every summary. Biometric trends from the ring show alongside the text. Session booking works from a dropdown of available dates. One-tap crisis trigger activates automatically when biometrics and emotional trends cross the threshold.

2.2 Psychologist Portal

Clinicians see only their assigned patients, sorted by crisis priority. AI-synthesized clinical notes in OAP format save hours of documentation time. Follow-up tasks get assigned and graded with proof-based completion - patients upload evidence, psychologists score it. Booking management with accept/waitlist. Encrypted session notes. The AI sidebar gives pre-session briefs, cross-patient pattern detection, and alerts when a patient has been silent too long.

2.3 Adaptive Smart Room - Shared Sensory Modulation

The smart room adjusts lighting (amber shift for calming, cool for alertness), sound (low-frequency pink noise at 60-200 Hz), and scent (lavender, vanilla) based on real-time physiological state. Crucially, it reads both the patient and the psychologist simultaneously. When both are elevated, the environment modulates for the dyad - creating a shared calming field rather than leaving the clinician to regulate alone.

The current configuration is a prototype foundation. The architecture supports scaling from a single biometric ring with a web dashboard to a full sensory clinic room. Future iterations use a plugin system for additional sensors and actuators.

2.4 Crisis Engine

When a patient hits the emergency trigger, three things happen on a timer: 0-29 seconds: local audiovisual siren on the patient dashboard. At 30 seconds: the trusted contact gets an email with an acknowledgment link - no login required, just click to confirm. At 60 seconds: helpline escalation if nobody has acknowledged yet. If the psychologist acknowledges at any stage, everything stops and we record resolution time. The halt is instantaneous.

3. AI AS ASSISTANT, NOT REPLACEMENT

The AI never makes clinical decisions. Full stop. It does not diagnose, does not prescribe, does not override anything a human decides. Its role is strictly supportive.

The custom sentinel model (fine-tuned 7.2B parameters via Ollama) produces empathy-toned reflections for patients and structured clinical notes for psychologists. A TF-IDF emotion classifier over 28 GoEmotions labels tags journal text. Echo detection prevents the AI from just parroting the patients own words back at them - a surprisingly common failure mode in off-the-shelf models.

If a clinic needs absolute data privacy, the model runs entirely offline via Ollama - no data leaves the building. Cloud deployments can use Groq API for speed. Three-tier fallback ensures you never get a blank response: Ollama local first, Groq remote second, rule-based extraction last. The crisis engine and discrepancy classifier never touch any external API.

4. SCIENTIFIC FOUNDATION - BIOLOGY MEETS PSYCHOLOGY

Sentinel's assessment approach works because the body tells the story before the mind does. Research shows elevated resting heart rate and disrupted sleep precede self-reported emotional deterioration by hours or days. By combining biometric and emotional streams into a single pipeline, you catch the signal earlier.

Three data layers feed the engine: Biometric data from the ring - heart rate, stress levels, sleep duration, SpO2, mood - deterministically seeded per user per hour. Emotional data from journal text, analyzed across 28 emotion labels via a TF-IDF + LogisticRegression classifier. A discrepancy detector that flags mismatches between what the body says and what the text says - because a patient writing I am fine with a heart rate of 120 is not fine.

The emotion classifier runs locally as a 4 MB Python pickle. No cloud dependency. Deterministic inference. Zero neural black boxes.

5. SELF-CARE FOR THE CLINICIAN

The psychologist dashboard includes self-monitoring metrics - cumulative session load, average escalation response time, and passive biometric trends from their own ring if they wear one. This is not surveillance. It is awareness. Secondary traumatic stress and fatigue creep up slowly. A dashboard that shows you the data helps you catch it before it catches you.

6. ACCESSIBILITY & DEPLOYMENT

The entire platform runs on free-tier infrastructure. Ollama handles local inference - no GPU required, runs on any machine with 8 GB RAM. Groq handles cloud fallback. The backend is FastAPI on uvicorn, frontend is React with Vite, both containerized and deployable with a single docker-compose up. SQLite in WAL mode replaces the earlier JSON storage, giving us transaction-safe operations across 15 database tables.

Zero financial barrier for any clinic, school, or community center. No licensing fees. No per-seat pricing. No vendor lock-in. Everything runs on your own hardware or a free Render instance.